

Innovation Summary

Project Title: SymbioLink AI

Problem Statement: MSMEs inside a shared industrial cluster routinely generate waste that could serve as a nearby unit's raw material, but no digital system exists to discover and act on such matches. Usable material goes to landfill, and exchanges depend on personal networks rather than a searchable market.

Proposed Solution: SymbioLink AI models an industrial cluster as a graph and finds direct matches and 2-3 hop exchange chains between units, ranked by savings, distance, and trust. Listings are created through a web form or a WhatsApp/voice-note message, removing literacy and app-adoption barriers. A predictive layer forecasts upcoming waste, and pooling batches small loads together.

Use of Artificial Intelligence: A trained TF-IDF and logistic-regression classifier parses free-text and voice listings; a Ridge regressor forecasts a unit's next waste listing; a logistic-regression ranker scores match chains by saving, distance, trust, and similarity; a Random Forest classifies waste type from photos. Every prediction carries a plain-language explanation, keeping the system transparent rather than a black box.

Innovation/Novelty: Multi-hop symbiosis matching, pre-emptive predictive listings, IoT-sensor auto-listing, GPS-verified logistics, and an in-platform carbon-credit marketplace linked to a live BRSR ESG report.

Target Users/Beneficiaries: MSME owners in manufacturing clusters, cluster associations, and sustainability/ESG officers.

Expected Impact: Reduced landfill waste, lower input costs for MSMEs, measurable CO₂e savings, and audit-ready ESG reporting at cluster scale.

Current Development Stage: Working Flask/MySQL prototype with three trained ML models across eleven integrated modules, on a synthetic dataset modeled on documented Indian industrial clusters; pre-pilot.